

CoFiTok: Coarse-to-Fine Denoising Tokens for Pixel-Space Diffusion

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Abstract

Pixel-space diffusion models usually predict a full dense noise field with a single monolithic network output. We introduce CoFiTok, a denoising-token factorization that represents the dense noise prediction as an ordered sum of compressed negative-noise components. Each token is expanded by a restricted, condition-free synthesis operator that only sees the current token, so prefix sums form valid partial noise predictions and produce controllable partial denoising results in pixel space. Across eight current datasets spanning CIFAR-10, ImageNet-family 64x64 and 256x256 settings, FFHQ-64, and AFHQv2-64, CoFiTok improves cross-batch prefix path AUC over an endpoint-only factorized control on all datasets and avoids zero-token leakage that appears when the synthesis operator is made deeper. In a two-seed ImageNet-256 confirmatory study, the learned order ranks first among all 24 K4 permutations for both seeds while preserving the endpoint sum. Frechet-style generation metrics favor EDM or direct dense epsilon on most rows, so the supported claim is not unconditional generation superiority; it is that ordered restricted dense-noise factorization yields prefix-controllable denoising without relying on a VAE-style decoder.

Introduction

Diffusion models reverse a noising process by repeatedly predicting a dense noise field. This dense prediction is effective, but it is also monolithic: there is no direct way to ask how much of the denoising update has been explained, whether the update has an interpretable coarse-to-fine order, or how to stop after a valid partial prediction. Ordered visual tokenizers and diffusion-autoregressive methods suggest that generation can benefit from structured intermediate representations. The question in this paper is narrower: can the dense pixel-space noise prediction itself be factorized into ordered denoising tokens?

CoFiTok rewrites the predicted noise field as

$$\hat{\epsilon}^{(K)} = \sum_{k=1}^K S_k \left(z^{(k)} \right), \quad (1)$$

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where each $z^{(k)}$ is a compressed negative-noise token and S_k expands only that token into one dense component. Prefixes are meaningful by construction:

$$\hat{\epsilon}^{(1:m)} = \sum_{k=1}^m S_k \left(z^{(k)} \right), \quad (2)$$

$$\hat{x}_0^{(m)} = \frac{x_t - \sigma_t \hat{\epsilon}^{(1:m)}}{\alpha_t}. \quad (3)$$

The central design constraint is that the synthesis operator must not become a decoder. In the default CoFiTok model, S_k is token-only, bias-free, shallow, local, and condition-free. It does not receive the noised image, timestep, class label, text prompt, previous tokens, skip features, or a learned constant. This restriction forces the token itself to carry the sample-specific negative-noise information and gives the zero-token test a concrete meaning: $S_k(0) = 0$ by construction.

The current evidence supports a scoped method claim. CoFiTok learns ordered denoising components across eight datasets, including ImageNet-family 64x64 and 256x256 settings and non-classification FFHQ/AFHQv2 rows. Compared with an endpoint-only factorized predictor, CoFiTok consistently improves prefix/path alignment, while matched objective, prediction, and synthesis ablations expose why ordering and restricted synthesis matter. Generated-sample distribution metrics do not show a stable unconditional generation-quality win, which clarifies the present contribution: CoFiTok makes dense noise prediction prefix-controllable without relying on a final VAE-style decoder.

Contributions.

1. We formulate dense pixel-space diffusion noise prediction as an ordered sum of compressed denoising-token components.
2. We introduce a restricted, condition-free synthesis operator whose token-only interface and bias-free construction prevent decoder-style shortcuts and make $S_k(0) = 0$ testable.
3. We train CoFiTok with a denoise-path objective that makes token prefixes approximate a partial denoising trajectory rather than only the final clean image.
4. We evaluate prefix controllability, component ordering, non-degeneration, endpoint quality, matched short-run

generation metrics, and tokenizer-related reconstruction baselines across the current eight-dataset evidence matrix.

Related Work

Diffusion models and latent diffusion. Denoising diffusion probabilistic models learn to reverse a Gaussian noising process by predicting noise targets (Ho, Jain, and Abbeel 2020). Latent diffusion models move this process into a learned latent space and decode the final latent through a VAE-style decoder (Rombach et al. 2022). CoFiTok instead stays in the pixel-space denoising algebra and factorizes the dense noise prediction itself. The CoFiTok token is not a final image latent; it is a compressed negative-noise component that directly contributes to the pixel-space denoising update.

Trajectory ordering and latent-pixel interaction. Latent Forcing studies how changing the schedule between latent and pixel variables changes the information revealed by the diffusion trajectory (Baade et al. 2026). CoFiTok is inspired by this view of trajectory ordering, but tokenizes the noise prediction inside the reverse trajectory rather than introducing a separate latent-pixel trajectory.

Ordered visual tokenizers. Spectral Image Tokenizer, FlexTok, Matryoshka Representation Learning, M3, and related nested-token approaches organize image or multimodal representations so that prefixes or subsets can be useful (Esteves, Suhail, and Makadia 2025; Bachmann et al. 2025; Kusupati et al. 2024; Cai et al. 2024). These methods operate on image representations or visual-token budgets. CoFiTok operates on dense diffusion noise prediction. Its prefixes are valid pixel-space denoising updates, not partial decodes from a general image tokenizer.

Diffusion-autoregressive visual tokens. Selftok and D-AR are the closest conceptual neighbors because they connect diffusion processes and autoregressive tokens (Wang et al. 2025; Gao and Shou 2025). CoFiTok differs in the represented object and in the anti-degeneration mechanism. CoFiTok tokens are continuous compressed negative-noise components, and the synthesis operator is deliberately weak and condition-free. The main claim is not that diffusion steps are visual tokens, but that dense noise prediction can be decomposed into ordered components whose prefixes are directly usable in the diffusion denoising formula.

Method

Forward Noising and Dense Prediction

For a clean image x_0 , timestep t , and Gaussian noise ϵ , the forward process is

$$x_t = \alpha_t x_0 + \sigma_t \epsilon. \quad (4)$$

A conventional diffusion model predicts $\epsilon_\theta(x_t, t, c)$. CoFiTok instead predicts the ordered component sum in Eq. 1. The current implementation uses unconditional models for the main MVP evidence. Class labels may exist in datasets, but the restricted synthesis operator never receives labels,

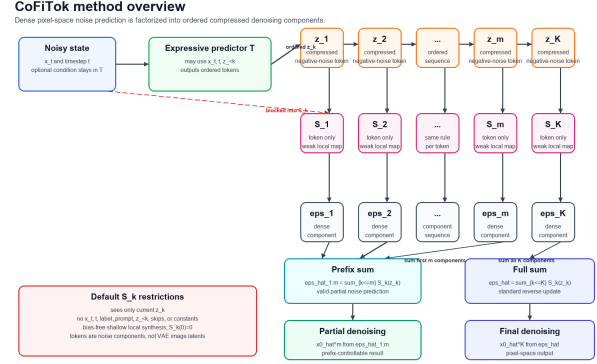


Figure 1: CoFiTok factorizes dense pixel-space diffusion noise prediction into ordered compressed denoising tokens. The predictor may be expressive, but each synthesis operator S_k receives only the current token and is restricted to shallow, condition-free, bias-free local synthesis. Any prefix of components forms a valid partial noise prediction and therefore a valid pixel-space denoising result. Dashed blocked paths mark information that must not enter S_k , including x_t , timesteps, labels, prompts, previous tokens, skip features, and learned constants.

prompts, timesteps, previous tokens, or image features. Figure 1 shows the dataflow and marks the blocked paths into S_k that define the default method boundary.

Next Denoising Tokens

The token predictor T_k may be expressive. Our default implementation sees the noised image, timestep, and previous tokens:

$$z^{(1)} = T_1(x_t, t), \quad (5)$$

$$z^{(k)} = T_k(x_t, z^{(<k)}, t), \quad k > 1. \quad (6)$$

In the tiny MVP model, T_k is a small U-Net-like predictor with token heads. Autoregressive feedback is not part of the synthesis restriction or the core factorization claim: a simultaneous variant may instead use $z^{(k)} = T_k(x_t, t)$. We evaluate this choice directly.

Restricted Synthesis Operator

Each token is expanded independently:

$$\epsilon^{(k)} = S_k(z^{(k)}). \quad (7)$$

The default S_k is a bias-free projection followed by a bias-free local convolution. It is token-only and condition-free. Because no layer has bias and there is no learned constant input, zero tokens map to zero components:

$$S_k(0) = 0. \quad (8)$$

This restriction separates CoFiTok from a learned decoder. A deep- S_k ablation is included only to test degeneration risk: it uses biased nonlinear convolutions while preserving the token-only interface. This ablation can improve apparent endpoint metrics but fails the zero-token diagnostic, so it is not the default method.

Prefix Denoising

For any prefix length m , CoFiTok forms the prefix prediction and pixel-space denoising result in Eq. 3. Thus every prefix is a valid partial noise prediction and every prefix produces a pixel-space denoising result. This is the primary interface exposed by the method.

Denoise-Path Objective

The current MVP objective is

$$\mathcal{L} = \mathcal{L}_\epsilon + 0.15 \mathcal{L}_{\text{path prefix}} + 0.30 \mathcal{L}_{\text{path component}}. \quad (9)$$

The denoise path interpolates from the zero-update x_t reconstruction toward coarse-to-fine spatial targets. The progress schedule uses power 1.5 in the current best MVP configuration. This objective deliberately avoids forcing early prefixes to be immediately clean; instead, prefixes are trained to follow a partial denoising path.

Experiments

Setup

Datasets and models. We evaluate on CIFAR-10, Tiny ImageNet-200, ImageNet-1K 64x64 HF, strict-source downsampled ImageNet-64, FFHQ-64, AFHQv2-64, ImageNet-256 10%, and full ImageNet-256. The HF ImageNet-64 row and the strict-source downsampled ImageNet-64 row are reported separately; older HF-derived results are not silently renamed as strict-source results. All matched experiments use small unconditional models and short training budgets. The primary model is K8 light CoFiTok at 64px and K4 at 256px. We compare it with a parameter-matched direct dense predictor, an endpoint-only factorized control, explicit objective/prediction/synthesis ablations, Improved DDPM, and EDM. FlexTok/TiTok reconstruction and official D-AR/MAR/ReTok 50K results are protocol-separated secondary context.

Protocols. The P0 rows match dataset, resolution, split, and nominal optimizer steps; batch size, nominal images seen, parameter count, and NFE are reported because the external methods are not compute matched. The direct dense control is the strictest factorization control: it is parameter matched and uses the same training budget, but is not claimed to be FLOP or latency matched. The broad matrix uses locked per-dataset evaluation sizes. Repeated and ablation claims use seeds 103/139 and 1,024 validation images at fixed $t = 500$. All power-sweep models are evaluated against the same progress-1.5 path.

Metrics and evidence tiers. We report endpoint PSNR and MSE, path AUC, effective token count, and zero-input energy. Lower path AUC means better alignment of all prefixes to the prescribed partial-denoising path; endpoint metrics alone do not measure that interface. Frechet-style values are internal comparison metrics rather than SOTA FID claims. The locked evidence matrix contains 80 matched P0 cells, 16 tokenizer eval-only cells, and 24 explicitly protocol-blocked cross-task cells; full settings and provenance are in the supplement.

Results

All-dataset evidence. Table 1 isolates the supported claim. C is CoFiTok, D is the parameter-matched direct dense predictor, and E is the endpoint-only factorized control. CoFiTok has lower path AUC than E on all eight datasets and exactly preserves the restricted zero-token contract. Its endpoint PSNR is close to, but not uniformly better than, direct dense.

Endpoint quality and prefixes. At the same 20k training budget, K8 light CoFiTok nearly matches the endpoint-only factorized control in endpoint x_0 MSE at $t = 500$ while greatly improving path alignment.

Across the four paired dataset-seed comparisons, CoFiTok lowers path AUC in 4/4 cases, with a mean relative reduction of 96.18%. This structural gain has a measurable endpoint tradeoff: endpoint x_0 MSE increases by 4.20% on average and is not lower in any of the four pairs. Figure 2 shows the corresponding qualitative prefix grids. In the full-validation deterministic sweep, K8 light also improves MSE AUC and low-res Frechet proxy AUC on both Tiny ImageNet-200 and ImageNet-64 HF, while the endpoint-only factorized control keeps the lower endpoint MSE.

Learned order and ImageNet-256 scaling. Order ablations keep the same component sum at the endpoint but change the prefix trajectory. This makes order evaluation a direct test of whether prefixes are organized rather than an endpoint-quality test.

Table 3 gives the preregistered ImageNet-256 K4 20k two-seed test. Besides a fixed random permutation and reverse order, it exhaustively evaluates all 24 component permutations over 1,024 images and reports a paired-bootstrap confidence interval for the mean of the 23 non-identity orders. Because every order sums the same components, the endpoint is preserved; the test isolates the organization of the prefix path.

The preregistered confirmatory test passes all seven gates. For both seeds, the learned order ranks 1st of 24, every non-identity permutation has worse path AUC, and the paired-bootstrap lower bounds for non-identity minus ordered AUC are positive (0.2489 and 0.2400). The endpoint component sum is invariant up to 3×10^{-15} MSE. Relative to the parameter-matched direct dense control, CoFiTok also lowers endpoint x_0 MSE by 6.73% on average in this ImageNet-256 confirmatory setting.

Generation and related-method context. Cross-batch validation confirms that endpoint-only factorization remains a strong endpoint baseline. Generated-sample metrics do not establish stable unconditional superiority: the matched controls favor endpoint-only on Tiny ImageNet-200 and ImageNet-64 HF under the internal streamed protocol, and EDM or direct dense wins the broad Frechet-style rows. Official D-AR/MAR/ReTok 50K and FlexTok/TiTok reconstruction results remain secondary protocol tiers in the supplement; none is used as a matched CoFiTok superiority claim.

Ablations

Matched component protocol. We isolate the K8 method components on ImageNet-64 HF with identical 5k-step train-

Table 1: Diagnostics for the supported CoFiTok claim. Higher PSNR is better; lower path AUC and zero ratio are better.

Dataset	C PSNR	D PSNR	C path	E path	C effK	E effK	C zero	deep- S_k zero
CIFAR-10	14.245	14.407	0.0967	0.7906	6.84	5.24	0.0000	0.0619
Tiny-IN	14.348	14.453	0.0731	1.1177	6.80	5.57	0.0000	0.0505
ImageNet-64 HF	14.897	15.004	0.0672	1.1120	6.79	5.33	0.0000	0.0522
ImageNet-64 strict	14.521	14.656	0.0705	1.0502	6.82	5.36	0.0000	0.0535
FFHQ-64	15.024	15.126	0.0676	1.0830	6.82	5.41	0.0000	0.0518
AFHQv2-64	15.167	15.154	0.0653	1.0185	6.82	5.45	0.0000	0.0561
ImageNet-256 10%	13.627	13.490	0.1416	1.5686	3.36	3.67	0.0000	0.0179
ImageNet-256	13.809	13.770	0.1325	1.5003	3.37	3.67	0.0000	0.0185

Table 2: Matched 20k two-seed comparison (mean $\pm$ sample standard deviation over seeds 103 and 139). Endpoint x_0 MSE and path AUC use 1,024 validation images at $t = 500$; lower is better. Higher is better for effective token count.

Dataset	Method	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
Tiny ImageNet-200	Endpoint-only factorized	0.1253 $\pm$ 0.0001	1.0098 $\pm$ 0.1430	5.312 $\pm$ 0.049	0.0000 $\pm$ 0.0000
Tiny ImageNet-200	CoFiTok	0.1296 $\pm$ 0.0015	0.0378 $\pm$ 0.0007	6.766 $\pm$ 0.015	0.0000 $\pm$ 0.0000
ImageNet-64 HF	Endpoint-only factorized	0.1096 $\pm$ 0.0006	0.9984 $\pm$ 0.1210	5.169 $\pm$ 0.108	0.0000 $\pm$ 0.0000
ImageNet-64 HF	CoFiTok	0.1151 $\pm$ 0.0016	0.0379 $\pm$ 0.0037	6.752 $\pm$ 0.012	0.0000 $\pm$ 0.0000

ing, seeds 103/139, and 1,024-image evaluation at $t = 500$. The full model uses $(\lambda_p, \lambda_c, p) = (0.15, 0.30, 1.5)$. Endpoint-only removes both path terms; the next two rows remove one term at a time; simultaneous prediction removes token feedback; deep synthesis replaces restricted S_k with a biased nonlinear decoder. Table 4 reports mean $\pm$ sample standard deviation.

Objective, prediction, and synthesis components. Relative to endpoint-only factorization, the full objective lowers mean path AUC by 93.0% with a 2.0% endpoint-MSE cost. Relative to removing only the prefix or component path term, it lowers path AUC by 45.7% and 37.9%, respectively, while endpoint MSE changes by less than 0.2%. Both path terms therefore improve the claimed interface rather than endpoint fit. Sequential feedback is a modest implementation gain, not a core requirement: full improves the two-seed mean by 2.8% in path AUC and 1.0% in endpoint MSE over simultaneous heads, but wins path AUC on only one seed. Deep synthesis attains lower path AUC yet has zero ratio 0.0485 ± 0.0050 versus exactly zero for restricted S_k ; its apparent gain violates the identifiability contract. Figure 2 visualizes zero/shuffle/deep- S_k diagnostics.

Hyperparameter sensitivity. Figure 3 varies the two path-loss weights, progress power, and token count. Every point is a two-seed mean with sample standard-deviation bars; the dashed line marks the default. Crucially, the power sweep is evaluated against one common progress-1.5 path rather than allowing each training target to define its own metric. Raising either path weight steadily improves path AUC with little endpoint movement; $(0.15, 0.30)$ is a moderate tradeoff rather than the path-only optimum. Power 1.5 is the clear

minimum under the common target. K8 gives the lowest mean path AUC, while K4 has slightly lower endpoint MSE and K16 adds cost without a 5k-step path gain.

Component-separation stress test. We treat component decorrelation as a pressure-term ablation rather than a default objective. The headline CoFiTok rows use the no-decorrelation light denoise-path objective because it most directly measures ordered prefix denoising. On Tiny ImageNet-200, a conservative schedule (weight=0.003, start step 3000, warmup 1500, full weight at step 4500 of 5000) reduces component correlation on two seeds while keeping the ordered-prefix penalty small.

Across the two seeds, scheduled decor 0.003 changes ordered path AUC by +0.0036, mean absolute component cosine by -0.0328, final MSE by -0.0002, and effective token count by +0.0336. Thus mild separation pressure can make components less correlated without materially breaking ordered prefix behavior. Stronger decor weights reduce cosine further but impose larger path-AUC penalties, so we keep them as stress-test ablations rather than the default CoFiTok objective.

Discussion

CoFiTok suggests that dense pixel-space diffusion noise prediction can be treated as a sequence of ordered denoising-token components. The restricted synthesis operator is central: it gives the prefix interface meaning and prevents the method from becoming a generic latent decoder. The denoise-path objective is also central because early prefixes should represent partial denoising rather than clean reconstructions.

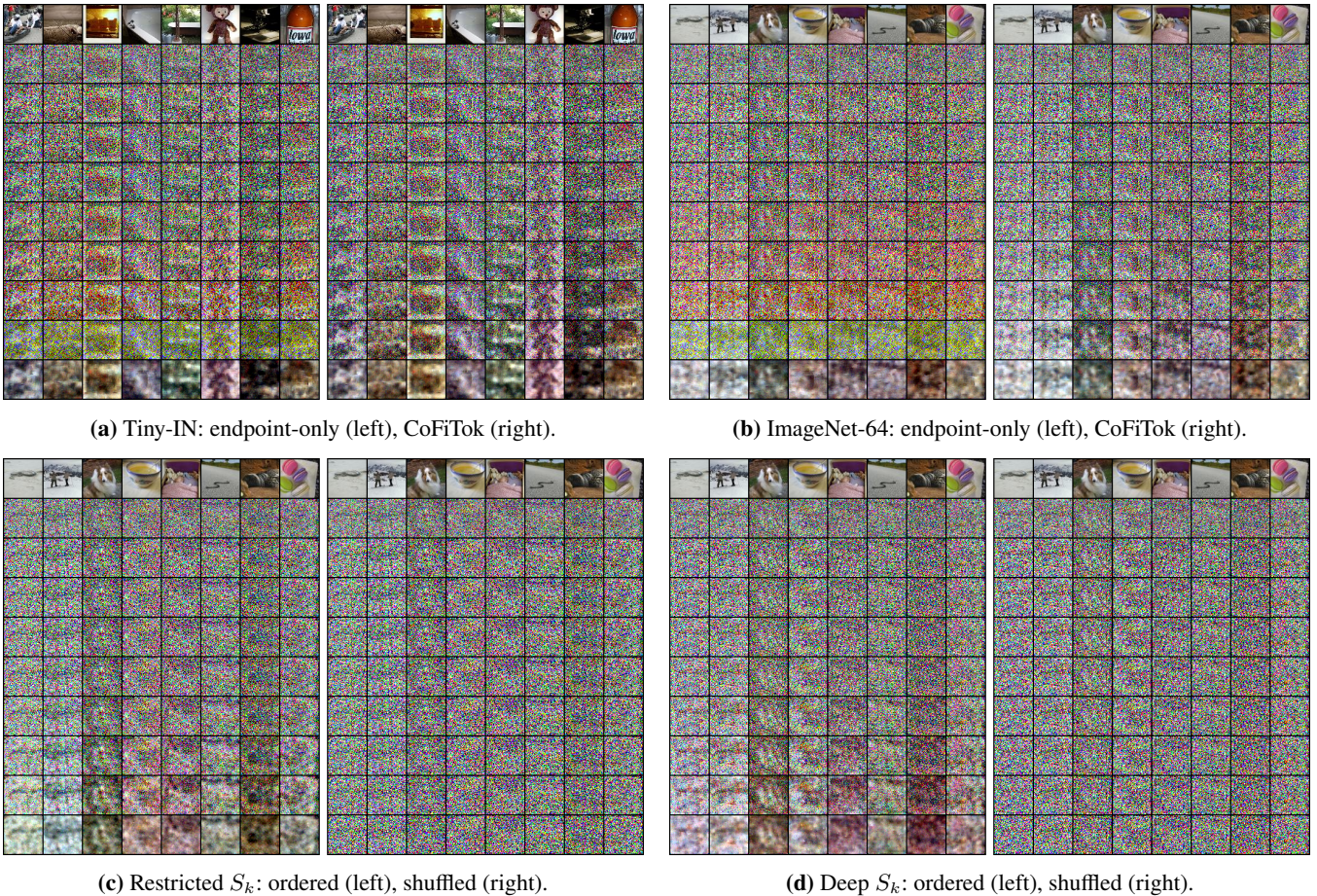


Figure 2: Qualitative prefix and degeneration evidence assembled from four standalone subfigure files. In (a–b), each half shows the same examples across the prefix path; CoFiTok produces more organized intermediate denoising than endpoint-only factorization. In (c–d), shuffling breaks sample alignment, while deep synthesis can carry structure that the restricted, zero-preserving operator cannot. Quantitative path and zero-token results are reported in Tables 1 and 4.

The strongest empirical result is the all-dataset prefix-control diagnostic: CoFiTok has lower cross-batch path AUC than the endpoint-only factorized control on all eight datasets, while matched objective, prediction, order, and deep- S_k ablations identify the roles of path supervision and restricted synthesis. The most important negative result is that generated-sample Frechet-style metrics do not show a stable unconditional generation-quality win. Together, these results define the current scope of the contribution.

Limitations

The current models are small and trained with short MVP budgets. Frechet-style sample-quality results are mixed and do not show a robust CoFiTok generation win; EDM is stronger on most generation rows, while direct dense prediction provides the strict factorization control. P1 FlexTok and TiTok rows use official pretrained reconstruction at 256x256 and are not retrained generation baselines. D-AR, MAR, and ReTok use separate official ImageNet-256 eval-only protocols and remain protocol-blocked under the

matched-dataset/step training definition. Seed repeats cover the main endpoint-only/K8 comparison but not every ablation. The AAAI-27 adaptation is an anonymous submission build and still needs author metadata and final checklist decisions before any camera-ready version. Sequential token prediction can also cost more latency and memory than a direct dense head; the current audit reports parameters, batch, nominal training images, and NFE, but not a standardized cross-implementation FLOP or wall-clock benchmark.

Conclusion

CoFiTok factorizes pixel-space diffusion noise prediction into an ordered sequence of compressed denoising tokens. Each token is expanded by a restricted, condition-free synthesis operator into a dense negative-noise component, and prefix sums yield valid partial denoising results. Across the current eight-dataset evidence matrix, CoFiTok consistently improves prefix-control diagnostics over an endpoint-only factorized control and passes the restricted zero-token diagnostic where deep synthesis leaks token-independent

Table 3: ImageNet-256 K4 20k confirmatory order and endpoint results over 1,024 validation images at $t = 500$.

Seed	E path	C ordered	non-ID mean	reverse	rank/24	Δ CI low	D endpoint	C endpoint
103	1.7879	0.0664	0.3157	0.7192	1	0.2489	0.1234	0.1133
139	1.4401	0.0678	0.3082	0.6821	1	0.2400	0.1244	0.1179

Table 4: Matched two-seed component ablations on ImageNet-64 HF. Lower is better for endpoint MSE, path AUC, and zero ratio; higher effective K indicates broader component use. All path values use the same progress-1.5 target.

Variant	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
CoFiTok (full)	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015	0.0000 ± 0.0000
Endpoint-only factorized	0.1275 ± 0.0019	0.9991 ± 0.1607	5.366 ± 0.054	0.0000 ± 0.0000
No path-prefix term	0.1300 ± 0.0003	0.1269 ± 0.0096	6.547 ± 0.027	0.0000 ± 0.0000
No path-component term	0.1302 ± 0.0012	0.1105 ± 0.0013	6.752 ± 0.069	0.0000 ± 0.0000
Simultaneous tokens	0.1314 ± 0.0016	0.0709 ± 0.0081	6.829 ± 0.025	0.0000 ± 0.0000
Deep synthesis	0.1496 ± 0.0305	0.0463 ± 0.0051	6.774 ± 0.032	0.0485 ± 0.0050

structure. The current evidence therefore supports a scoped contribution about ordered restricted dense-noise factorization, while leaving raw unconditional generation quality to stronger monolithic diffusion baselines. The next step is to scale the protocol with stronger backbones, longer training, and broader seed coverage.

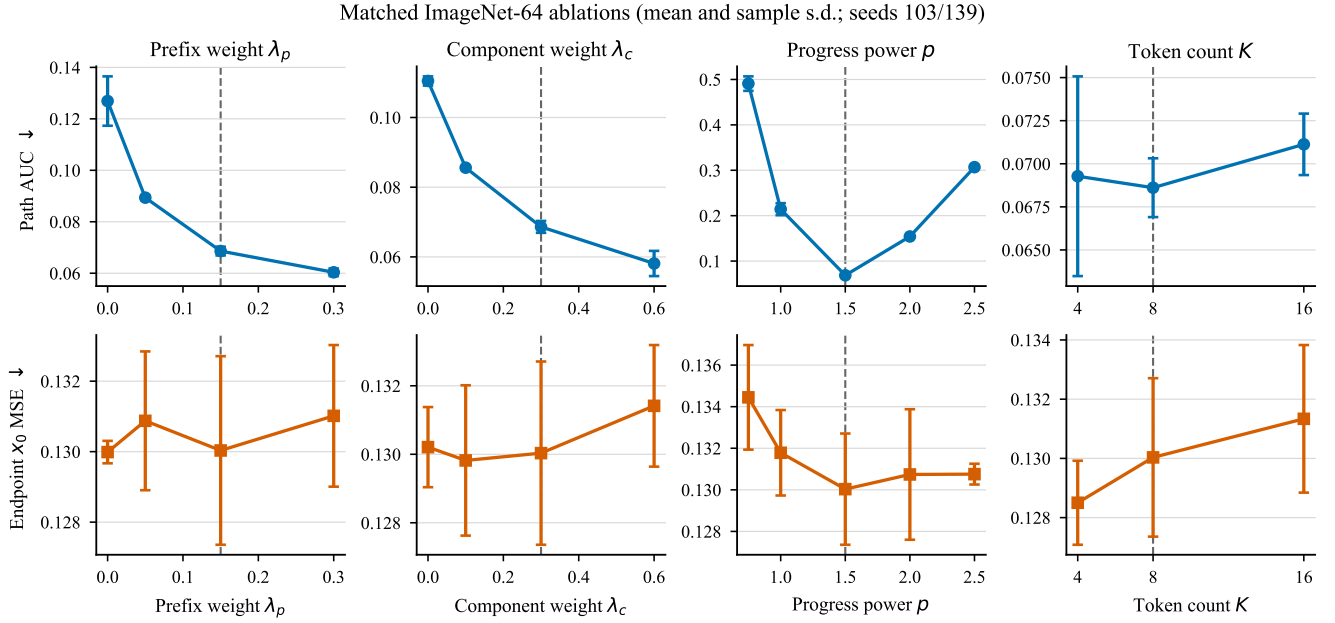


Figure 3: Matched ImageNet-64 HF hyperparameter ablations. Top: denoise-path AUC; bottom: endpoint x_0 MSE. Curves vary prefix weight λ_p , component weight λ_c , progress power p , and token count K . Lower is better; bars show sample standard deviation over seeds 103/139.

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